

Discrete Mathematics (DM)

GATE Computer Science & IT Previous Year Questions (2016 – 2025) | Total Questions: 15

Question 1 of 15 | GATE 2025 Set 1 (CS1) | Q.48 (Computer Science & IT)

Q.48

Which of the following predicate logic formulae/formula is/are CORRECT representation(s) of the statement: "Everyone has exactly one mother"?

The meanings of the predicates used are:

- mother(y, x): y is the mother of x
- noteq(x, y): x and y are not equal

- (A) $\exists x \exists y \exists z (\text{mother}(y, x) \wedge \neg \text{mother}(z, x))$
- (B) $\exists x \exists y [\text{mother}(y, x) \wedge \exists z (\text{noteq}(z, y) \rightarrow \neg \text{mother}(z, x))]$
- (C) $\exists x \exists y [\text{mother}(y, x) \rightarrow \exists z (\text{mother}(z, x) \wedge \neg \text{noteq}(z, y))]$
- (D) $\exists x \exists y [\text{mother}(y, x) \wedge \neg \exists z (\text{noteq}(z, y) \wedge \text{mother}(z, x))]$

Question 2 of 15 | GATE 2025 Set 1 (CS1) | Q.49 (Computer Science & IT)

Q.49

$A = \{0, 1, 2, 3, \dots\}$ is the set of non-negative integers. Let F be the set of functions from A to itself. For any two functions, $f_1, f_2 \in F$, we define

$$(f_1 \oplus f_2)(n) = f_1(n) + f_2(n)$$

for every number n in A . Which of the following is/are CORRECT about the mathematical structure $(F, \oplus)$?

- (A) $(F, \oplus)$ is an Abelian group.
- (B) $(F, \oplus)$ is an Abelian monoid.
- (C) $(F, \oplus)$ is a non-Abelian group.
- (D) $(F, \oplus)$ is a non-Abelian monoid.

Question 3 of 15 | GATE 2025 Set 2 (CS2) | Q.42 (Computer Science & IT)

Q.42

Let F be the set of all functions from $\{1, \dots, n\}$ to $\{0, 1\}$. Define the binary relation $\square$ on F as follows:

$\square f, g \in F, f \square g$ if and only if $\exists x \in \{1, \dots, n\}, f(x) \leq g(x)$, where $0 \leq 1$.

Which of the following statement(s) is/are TRUE?

- (A) $\square$ is a symmetric relation

- (B) $(F, \sqsubseteq)$ is a partial order
- (C) $(F, \sqsubseteq)$ is a lattice
- (D) $\sqsubseteq$ is an equivalence relation

Question 4 of 15 | GATE 2024 Set 1 (CS1 S5) | Q.51 (Computer Science & IT)

Q.51

The chromatic number of a graph is the minimum number of colours used in a proper colouring of the graph. Let G be any graph with n vertices and chromatic number k . Which of the following statements is/are always TRUE?

- (A) G contains a complete subgraph with k vertices
- (B) G contains an independent set of size at least n/k
- (C) G contains at least $k(k-1)/2$ edges
- (D) G contains a vertex of degree at least k

Question 5 of 15 | GATE 2024 Set 2 (CS2 S6) | Q.34 (Computer Science & IT)

Q.34

Let P be the partial order defined on the set $\{1,2,3,4\}$ as follows

$$P = \{(x, y) \mid x \sqsubseteq y \text{ in } \{1,2,3,4\}\} \cup \{(1,2), (3,2), (3,4)\}$$

The number of total orders on $\{1,2,3,4\}$ that contain P is _____

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 6 of 15 | GATE 2023 Single Session | Q.48 (Computer Science & IT)

Q.48

Let $f : A \rightarrow B$ be an onto (or surjective) function, where A and B are nonempty sets. Define an equivalence relation $\sqsim$ on the set A as

$$a_1 \sqsim a_2 \text{ if } f(a_1) = f(a_2),$$

where $a_1, a_2 \in A$. Let $E = \{[x] : x \in A\}$ be the set of all the equivalence classes under $\sqsim$. Define a new mapping $F : E \rightarrow B$ as

$$F([x]) = f(x),$$

for all the equivalence classes $[x]$ in E .

Which of the following statements is/are TRUE?

- (A) F is NOT well-defined.
- (B) F is an onto (or surjective) function.
- (C) F is a one-to-one (or injective) function.
- (D) F is a bijective function. CS

Question 7 of 15 | GATE 2022 Single Session | Q.36 (Computer Science & IT)

Q.36

Which one of the following is the closed form for the generating function of the sequence

0

{

}

n

n

a

□

defined below?

1,

is odd

1,

otherwise

n

n

n

a

□

□

□□

□

(A) □ □ □ □ 2 2 2 1 1 1 1 x x x x □ □ □

(B) □ □ 2 2 2 (3) 1 1 1 x x x x □ □ □

(C) □ □ 2 2 2 1 1 1 x x x □ □ □

(D) 2 2 1 (1) 1 x x x □ □ □

Question 8 of 15 | GATE 2022 Single Session | Q.50 (Computer Science & IT)

Q.50

The following simple undirected graph is referred to as the Peterson graph.

Which of the following statements is/are TRUE?

(A) The chromatic number of the graph is 3.

(B) The graph has a Hamiltonian path.

(C) The following graph is isomorphic to the Peterson graph.

(D) The size of the largest independent set of the given graph is 3. (A subset of vertices of a graph form an independent set if no two vertices of the subset are adjacent.)

Question 9 of 15 | GATE 2019 Single Session | Q.10 (Computer Science & IT)

Q.10 Let G be an arbitrary group. Consider the following relations on G :

R_1 : $\forall a, b \in G, a R_1 b$ if and only if $\exists g \in G$ such that $a = g^{-1}bg$

R_2 : $\forall a, b \in G, a R_2 b$ if and only if $a = b^{-1}$

Which of the above is/are equivalence relation/relation(s)?

- (A) R_1 and R_2
- (B) R_1 only
- (C) R_2 only
- (D) Neither R_1 nor R_2 CS 3/15

Question 10 of 15 | GATE 2019 Single Session | Q.12 (Computer Science & IT)

Q.12 Let G be an undirected complete graph on n vertices, where $n > 2$. Then, the number of different Hamiltonian cycles in G is equal to

- (A) $n!$
- (B) $(n-1)!$
- (C) 1
- (D) $(n-1)! \cdot 2$

Question 11 of 15 | GATE 2018 Single Session | Q.1 (Computer Science & IT)

Q.1

Which one of the following is a closed form expression for the generating function of the sequence $\{a_n\}$, where $a_n = 2n + 3$ for all $n = 0, 1, 2, \dots$?

- (A) $\sum_{n=0}^{\infty} (2n+3)x^n$
- (B) $\sum_{n=0}^{\infty} (2n+3)x^{n+1}$
- (C) $\sum_{n=0}^{\infty} (2n+3)x^{n+2}$
- (D) $\sum_{n=0}^{\infty} (2n+3)x^{n+3}$

Question 12 of 15 | GATE 2018 Single Session | Q.18 (Computer Science & IT)

Q.18 The chromatic number of the following graph is _____.

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[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 13 of 15 | GATE 2018 Single Session | Q.43 (Computer Science & IT)

Q.43 Let G be a graph with $100!$ vertices, with each vertex labelled by a distinct permutation of

the numbers $1, 2, \dots, 100$. There is an edge between vertices u and v if and only if the label of u can be obtained by swapping two adjacent numbers in the label of v . Let y denote the degree of a vertex in G , and z denote the number of connected components in G .

Then, $y + 10z = \underline{\hspace{2cm}}$.

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[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 14 of 15 | GATE 2016 Set 2 (CS2) | Q.3 (Computer Science & IT)

Q.3

The minimum number of colours that is sufficient to vertex-colour any planar graph is

.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 15 of 15 | GATE 2016 Set 2 (CS2) | Q.27 (Computer Science & IT)

Q.27

Which one of the following well-formed formulae in predicate calculus is NOT valid?

- (A) $(\forall x p(x) \rightarrow \exists x q(x)) \rightarrow (\exists x \neg p(x) \rightarrow \exists x q(x))$
- (B) $(\forall x p(x) \rightarrow \exists x q(x)) \rightarrow \exists x (p(x) \rightarrow q(x))$
- (C) $\exists x (p(x) \rightarrow q(x)) \rightarrow (\exists x p(x) \rightarrow \exists x q(x))$
- (D) $\exists x (p(x) \rightarrow q(x)) \rightarrow (\exists x p(x) \rightarrow \exists x q(x))$ CS(Set B) 7/18 GATE 2016